COMPUTER NETWORK: PHYSICAL LAYER

ASSIGNMENT 1

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Outline

1 Multiple Choice Questions (MCQs)

MCQ Q1: OSI Model Layer Matching – Question

Question MCQ Q1

Question: Match the networking components with their corresponding OSI model layers.

Set A: P. Hub, Q. Bridge, R. Router, S. Server

Set B: i. Physical Layer, ii. Data Link Layer, iii. Network Layer, iv. Application Layer

Options:

(A) P-i, Q-ii, R-iii, S-iv

(B) P-ii, Q-ii, R-iii, S-iii

(C) P-ii, Q-iii, R-iii, S-iii

(D) P-i, Q-iii, R-iii, S-iv

MCQ Q1: OSI Model Layer Matching – Strategy

Problem-Solving Strategy

- Recall the OSI layer at which each networking device operates.
- Hub repeats electrical signals $\rightarrow$ Physical layer.
- Bridge filters based on MAC addresses $\rightarrow$ Data link layer.
- Router forwards based on IP addresses $\rightarrow$ Network layer.
- Server provides services $\rightarrow$ Application layer.

MCQ Q1: OSI Model Layer Matching – Solution

Solution

Answer: (A) P-i, Q-ii, R-iii, S-iv

Hub (P) $\rightarrow$ Physical layer (i), Bridge (Q) $\rightarrow$ Data Link layer (ii), Router (R) $\rightarrow$ Network layer (iii), Server (S) $\rightarrow$ Application layer (iv).

MCQ Q2: PDU Matching – Question

Question MCQ Q2

Question: Match the following communication layers with their data unit names.

Set A: A. Link Layer, B. Network Layer, C. Application Layer, D. Transport Layer

Set B: 1. Message, 2. Segment, 3. Packet/Datagram, 4. Frame

Options:

(A) A-3, B-4, C-2, D-1

(B) A-4, B-3, C-2, D-1

(C) A-4, B-3, C-1, D-2

(D) A-3, B-4, C-1, D-2

MCQ Q2: PDU Matching – Strategy

Problem-Solving Strategy

- Match each layer with its Protocol Data Unit (PDU).
- Link layer (Layer 2) packages data into **Frames**.
- Network layer (Layer 3) packages data into **Datagrams/Packets**.
- Application layer (Layer 5/6/7) deals with **Messages**.
- Transport layer (Layer 4) packages data into **Segments**.

MCQ Q2: PDU Matching – Solution

Solution

Answer: (C) A-4, B-3, C-1, D-2

Link Layer $\rightarrow$ Frame (4), Network Layer $\rightarrow$ Datagram (3), Application Layer $\rightarrow$ Message (1), Transport Layer $\rightarrow$ Segment (2).

MCQ Q3: Network Interconnection Device – Question

Question MCQ Q3

Question: What is the name of the device that enables communication between two dissimilar networks that use different protocol sets and topologies?

(A) Router

(B) Bridge

(C) Gateway

(D) Switch

MCQ Q3: Network Interconnection Device – Strategy

Problem-Solving Strategy

- Key phrase: "dissimilar networks" and "different protocols".
- Router connects similar networks with same protocols.
- Bridge connects same type networks at data link layer.
- Gateway performs protocol conversion between dissimilar networks.

MCQ Q3: Network Interconnection Device – Solution

Solution

Answer: (C) Gateway

Gateway is used between two dissimilar LANs and acts as a protocol converter, translating data packets from one protocol format to another.

MCQ Q4: Broadcast Traffic Device – Question

Question MCQ Q4

Question: Which networking device connects multiple systems in a network and forwards broadcast messages to all connected ports without any filtering?

(A) Switch

(B) Hub

(C) Modem

(D) Repeater

MCQ Q4: Broadcast Traffic Device – Strategy

Problem-Solving Strategy

- Consider each device's behavior when receiving broadcast traffic.
- Hub is a physical layer device that simply repeats signals out of all ports.
- It has no intelligence to analyze or filter frames.
- Therefore, it forwards all broadcasts.

MCQ Q4: Broadcast Traffic Device – Solution

Solution

Answer: (B) Hub

Every device connected to a hub is in the same broadcast domain. Whenever a transmission is received on any port, the hub broadcasts it to all other ports without any filtering.

MCQ Q5: OSI Layer Functions – Question

Question MCQ Q5

Question: Match the networking functions with the OSI layers where they primarily operate.

Set A: P. Path Selection, Q. Data Rate Management, R. Physical Medium Connection, S. User Service Access

Set B: 1. Data Link and Transport Layers, 2. Network Layer, 3. Application Layer, 4. Physical Layer

Options:

(A) P-1, Q-2, R-3, S-4

(B) P-1, Q-2, R-4, S-3

(C) P-2, Q-1, R-3, S-4

(D) P-2, Q-1, R-4, S-3

MCQ Q5: OSI Layer Functions – Strategy

Problem-Solving Strategy

- Route Selection is primary function of Network layer.
- Flow Control is handled by Data Link and Transport layers.
- Interface to transmission media is job of Physical layer.
- Access for end user is provided by Application layer.

MCQ Q5: OSI Layer Functions – Solution

Solution

Answer: (D) P-2, Q-1, R-4, S-3

Path Selection (P) $\rightarrow$ Network Layer (2), Data Rate Management (Q) $\rightarrow$ Data Link and Transport Layers (1), Physical Medium Connection (R) $\rightarrow$ Physical Layer (4), User Service Access (S) $\rightarrow$ Application Layer (3).

MCQ Q6: Network Topology Statements – Question

Question MCQ Q6

Question: Identify which statement about network topologies is incorrect.

(A) A ring network with N devices uses N-1 drop cables and N ring segments

(B) A bus network with N devices requires 1 drop line and N backbone cables

(C) A star network with N devices has N+1 connections and N ports

(D) All of the above statements are incorrect

MCQ Q6: Network Topology Statements – Strategy

Problem-Solving Strategy

- Analyze each topology claim.
- Ring: For N devices, you have N point-to-point links.
- Bus: For N devices, you need N drop lines, 1 backbone.
- Star: For N devices, you have N links and N ports.

MCQ Q6: Network Topology Statements – Solution

Solution

Answer: (D) All of the above statements are incorrect

Statement (A) is incorrect about ring topology. Statement (B) is incorrect about bus topology. Statement (C) is incorrect about star topology. Therefore, all statements are not true.

MCQ Q7: Networking Devices Description – Question

Question MCQ Q7

Question: Match each networking device with its correct description.
Set A: A. Translates between different protocols, B. Lacks processing intelligence, C. Hardware-only device without embedded programs, D. Amplifies and retransmits signals without analyzing them, E. Hardware device that runs software for decision making, F. Connects multiple LAN segments using MAC address filtering, G. Combines features of two specific devices
Set B: 1. Gateway, 2. Brouter, 3. Router, 4. Bridge, 5. Switch, 6. Repeater, 7. Hub

Options:

(A) A-1, B-7, C-5, D-6, E-3, F-4, G-2

(B) A-2, B-7, C-5, D-6, E-5, F-4, G-1

(C) A-2, B-4, C-5, D-6, E-5, F-7, G-1

(D) A-1, B-2, C-5, D-4, E-3, F-6, G-7

MCQ Q7: Networking Devices Description – Strategy

Problem-Solving Strategy

- Gateway $\rightarrow$ Protocol converter.
- Hub $\rightarrow$ Passive device, no intelligence.
- Switch $\rightarrow$ Pure electronic device with no software.
- Repeater $\rightarrow$ Retransmits with high signal.
- Router $\rightarrow$ Pure electronic with software.
- Bridge $\rightarrow$ Active device with filtering/forwarding.
- Brouter $\rightarrow$ Combination of bridge and router.

MCQ Q7: Networking Devices Description – Solution

Solution

Answer: (A) A-1, B-7, C-5, D-6, E-3, F-4, G-2

Gateway (A-1), Hub (B-7), Switch (C-5), Repeater (D-6), Router (E-3), Bridge (F-4), Brouter (G-2).

MCQ Q8: OSI Model Statements – Question

Question MCQ Q8

Question: Which statements about the OSI reference model are accurate?
(i) It breaks down network communication into manageable pieces, simplifying development and troubleshooting
(ii) It promotes interoperability between different manufacturers through standardization
(iii) It ensures modifications in one layer don't impact adjacent layers
(iv) Real-world protocol implementations rarely follow this model exactly
Options:

(A) Only (i) and (iv)

(B) Only (ii) and (iii)

(C) Only (i), (ii), and (iv)

(D) All four statements are correct

MCQ Q8: OSI Model Statements – Strategy

Problem-Solving Strategy

- Evaluate each statement against known benefits of OSI model.
- Layering simplifies design and troubleshooting.
- Standardization enables multi-vendor development.
- Layer independence allows quicker development.
- Real-world protocols (TCP/IP) don't exactly match OSI.

MCQ Q8: OSI Model Statements – Solution

Solution

Answer: (D) All four statements are correct

All statements accurately describe benefits and characteristics of the OSI model.

MCQ Q9: Best Effort Services – Question

Question MCQ Q9

Question: What features are NOT provided by best-effort delivery services like IP?

(A) Data verification

(B) Confirmation of packet delivery

(C) Data recovery

(D) All of the above

MCQ Q9: Best Effort Services – Strategy

Problem-Solving Strategy

- IP includes header checksum (error checking for header only).
- IP is unreliable $\rightarrow$ no datagram acknowledgment.
- IP has minimal error control $\rightarrow$ no error correction.
- Question asks what is NOT included.

MCQ Q9: Best Effort Services – Solution

Solution

Answer: (D) All of the above

While IP has header checksum (minimal error checking), it does not acknowledge datagrams or perform error correction. Therefore, all options are correct for what IP does NOT include.

MCQ Q10: Ethernet Efficiency – Question

Question MCQ Q10

Question: When does Ethernet achieve its highest efficiency?

(A) When both signal travel time and data transmission time are minimal

(B) When signal travel time is minimal but data transmission time is substantial

(C) When signal travel time is substantial but data transmission time is minimal

(D) When both signal travel time and data transmission time are substantial

MCQ Q10: Ethernet Efficiency – Strategy

Problem-Solving Strategy

- Efficiency formula: $Efficiency = \frac{1}{1+6.44a}$ where $a = \frac{t_p}{t_t}$.
- For high efficiency, 'a' must be low.
- 'a' is low when t_p (propagation delay) is low and t_t (transmission delay) is high.

MCQ Q10: Ethernet Efficiency – Solution

Solution

Answer: (B) When signal travel time is minimal but data transmission time is substantial

Efficiency $= 1/(1 + 6.44a)$ where $a = t_p/t_t$. For high efficiency, 'a' should be low, which occurs when propagation delay (t_p) is low and transmission delay (t_t) is high.